

BRYAN SANGWOO KIM

M.S. Student @ KAIST AI

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 South Korea, United States of America (Dual Citizenship)



RESEARCH INTERESTS

I have a broad interest in generative models that work across various modalities (images, video, text) and their applications. Much of my recent research involves using such multimodal systems to drive tangible solutions for practical, real-world challenges. **Keywords:** Multi-modal generative models, Image & video diffusion models, Inverse problems

EDUCATION

3/2024 - 2/2026	KAIST AI M.S., Artificial Intelligence (Advisor: Jong Chul Ye) GPA: 4.25/4.3	Seoul, South Korea
3/2018 - 2/2024	KAIST B.S., Computer Science, Biological Sciences (Double Major) Leave of Absence: Mandatory Korean Military Service (2021, 2022) GPA: 3.96/4.3 (Magna Cum Laude)	Daejeon, South Korea

PREPRINTS & PUBLICATIONS

PREPRINT First Author	Chain-of-Zoom: Extreme Super-Resolution via Scale Autoregression and Preference Alignment Bryan Sangwoo Kim , Jeongsol Kim, Jong Chul Ye Links:   
ICCV 2025 Co-First Author	FlowDPS: Flow-Driven Posterior Sampling for Inverse Problems Jeongsol Kim*, Bryan Sangwoo Kim *, Jong Chul Ye (*equal contribution) <i>IEEE/CVF International Conference on Computer Vision, 2025</i> Links:  
ICCV 2025 Second Author	Free²Guide: Training-Free Text-to-Video Alignment using Image LVLM Jaemin Kim, Bryan Sangwoo Kim , Jong Chul Ye <i>IEEE/CVF International Conference on Computer Vision, 2025</i> Links:  
CVPR 2025 Co-First Author	VideoGuide: Improving Video Diffusion Models without Training Through a Teacher's Guide Dohun Lee*, Bryan Sangwoo Kim *, Geon Yeong Park, Jong Chul Ye (*equal contribution) <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025</i> Links:   

AWARDS & HONORS

2024 - 2025	KAIST Support Scholarship (Graduate) Full Scholarship for Tuition & Stipend
2020	KAIST Department Honors Scholarship Department Valedictorian
2020 - 2023	National Excellence Scholarship (Natural Sciences and Engineering) Full Scholarship for Tuition
2018	KAIST Dean's List
2018 - 2024	KAIST Presidential Fellowship (KPF)
2018 - 2019	KAIST Support Scholarship (Undergraduate) Full Scholarship for Tuition
2016	Hansung Scholarship for Gifted Students

PROJECTS

- 7/2024 - Present **Acquisition of 3D Precise Information of Microstructure and Development of Authoring Technology for Ultra-high Precision Cultural Restoration** Project Researcher
Ministry of Culture, Sports and Tourism (South Korea)
: Developed a recursive resolution enhancement algorithm to achieve high resolutions in reconstructed CT images. Incorporated algorithms to further enhance consistency across multiple scales. Conducted experiments to enhance the CT image resolutions of various artifact data.
- 10/2024 - 3/2025 **Development of a PPG-based Respiratory Rate Prediction Algorithm** Project Leader
SkyLabs Co.
: Labeled respiratory rate segments using sensor data with peak detection algorithms. Developed and applied a template-matching SQL for data preprocessing. Developed a PPG foundation model with low-frequency VQVAE for time series tokenization and BERT for serial analysis. Trained a multi-token classifier with multiple token inputs and proved its effective prediction for a wide range of respiratory rates.
- 3/2024 - 8/2024 **Development of ECG-based Seizure Prediction, Detection, Post-Detection Models** Project Leader
SkyLabs Co.
: Developed an ECG foundation model with VQVAE for time series tokenization and BERT for serial analysis. Applied techniques (e.g., contrastive learning) to improve representations. Leveraged the ECG foundation model on the downstream tasks of seizure prediction/detection/post-detection by training classifiers on [CLS] token outputs. Final evaluation demonstrated 100% accuracy on seizure detection.

RESEARCH EXPERIENCE

- 6/2023 - 8/2023 **KAIST AI Research Internship (KAIRI)** Daejeon, South Korea
Bioimaging, Signal Processing, & Learning Lab, KAIST (Advisor: Prof. Jong Chul Ye)
: Studied/implemented diffusion models and their various applications, especially regarding inverse problems. Conducted some research into generative modeling of protein sequences.
- 3/2023 - 4/2023 **Research Service** Seoul, South Korea
Innerviz Co.
: Created a framework for accurate eye control (e.g., eye blink, gaze redirection) consisting of two distinct modules: a blink control module and a gaze redirection module. Used a novel data augmentation method to train each module, leveraging style mixing to obtain images with desired features. Work is organized and presented in the technical report 'EyeBAG: Accurate Control of Eye Blink and Gaze Based on Data Augmentation Leveraging Style Mixing' available on ArXiv. Link: [📄](#)
- 9/2020 - 2/2021 **Specified Research** Daejeon, South Korea
Systems Neuroscience Laboratory, KAIST (Advisor: Prof. Min Whan Jung)
: Analyzed oscillatory neural activities (ripples) in the mouse brain hippocampus with the Allen Mouse Brain Atlas dataset. Leveraged machine learning techniques to explore patterns and anomalies in mouse neural activities.
- 6/2019 - 12/2019 **Undergraduate Individual Study** Daejeon, South Korea
Brain x Machine Intelligence Lab, KAIST (Advisor: Prof. Sang Wan Lee)
: Learned machine learning basics and RL algorithms. Participated in development of software code for multi-voxel pattern analysis of fMRI neuroimaging data with MATLAB.

TEACHING EXPERIENCE & OUTREACH

- 7/2025 **Teaching Assistant @Samsung Financial Networks** Seoul, South Korea
Samsung Fire & Marine Insurance AI Business Expert Training Program
: Assisted hands-on workshop lectures on LLM basics, RAG, LangChain, and Streamlit prototyping.
- 8/2020 - 12/2020 **Teaching Assistant & Mentor @KSOP (KAIST Science Outreach Program)** Daejeon, South Korea
KAIST Science Outreach Program Online Academic Mentorship
: Provided online academic mentorship during the semester and intensive camp training during school break for socially disadvantaged students interested in mathematics and science.
- 8/2019 **Teaching Assistant & Mentor @Hansung Scholarship Summer Camp** Seoul, South Korea
6th Hansung Nobel Young Talent Scholarship Summer Camp
: Assisted a project on the transformation of traditional industries through blockchain technology. Participated as a mentor for high school students on the field of bioengineering.

OTHER EXPERIENCE

1/2020	Academic Collaboration Forum @Indian Institute of Technology, Madras <i>2020 KAIST-IITM Academic Collaboration Forum</i> : Participated in the 2020 KAIST-IITM Joint Research Challenge, organized between the KAIST Presidential Fellowship (KPF) group and the Centre for Innovation (CFI) group. Awarded 2nd place for proposing a solution to the challenge of 'Food Waste Management in Student Messes'.	Chennai, India
8/2019	Exploratory Research in Southeast Asia @Malaysia <i>Group Leadership Activity (GLA) Program</i> : Awarded 1st place in the Group Leadership Activity (GLA) program for interviewing experts in dementia-related work and sharing their messages with the world. Conducted additional exploratory research in the AACC (Atria-ADFM Community Corner) of Malaysia, fully funded by the KAIST Global Leadership Center.	Kuala Lumpur, Malaysia
6/2018 - 8/2018	Summer Session @University of California, Berkeley <i>2018 UC Berkeley Summer Session</i> : Participated in the UC Berkeley Summer Session, fully funded by the KAIST Global Leadership Center, privilege of the KAIST Presidential Fellowship (KPF). Attended courses 'Multivariable Calculus (Grade: A)' and 'Introduction to the History of the United States (Grade: A-)'. Global Research Frontiership @George Mason University <i>Global Research Frontiership (GRF)</i> : Participated in an outsourced research training program in the United States, organized by Daegu Science High School (DSHS). Attended a workshop on astrology at George Mason University (GMU).	Berkeley, CA, United States
6/2016		Fairfax, VA, United States
9/2015 - 7/2016	Global Research Program @Loudoun Academy of Science (AOS) <i>AOS/DSHS Research Collaboration</i> : Participated in a global research collaboration organized between Daegu Science High School (DSHS) and the Loudoun Academy of Science (AOS). Conducted research 'Synthesizing Hydrophobic POSS-modified Ethyl Cellulose' with fellow students from AOS.	Sterling, VA, United States

IT SKILLS

Programming Languages - Python, C, MATLAB
Programming Languages (Minor) - Java, Kotlin, C++, Rust
Technical Skills - HTML, JavaScript, CSS, LaTeX
Libraries - PyTorch, Tensorflow, NumPy, Pandas, Matplotlib, Scikit-learn

LANGUAGES

Korean - native, **English** - native

REFERENCES

3/2024 - Present	Jong Chul Ye M.S. Advisor (KAIST)	jong.ye@kaist.ac.kr
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